What makes a playlist successful?

Github

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Agenda

1. Exploring the data
2. Defining success
3. Understanding the drivers
4. Summary & next steps
5. Q&A

The dataset is sampled from available playlists on Spotify

Metadata

Owner
Playlist title
Genres
Moods
# of tracks, artists, albums

Engagement Metrics

Streams this month
Active Users stats over two months

427,000 playlists

The sample includes more than 400 thousand playlists where **411** owned by Spotify

100% US based

All playlist owners are based in the US

Likely a September snapshot

Inferred by title token count and retention ratios

26 Genres, 27 Moods

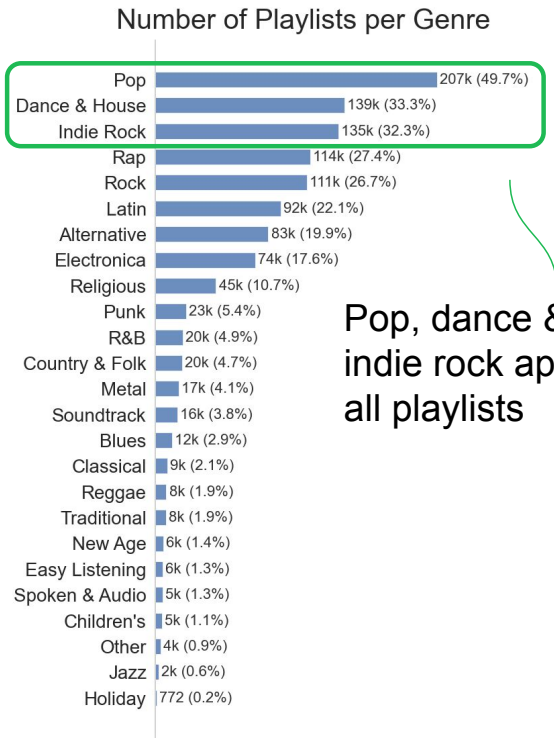
There are 26 distinct genres and 27 distinct moods across all playlists.

195 Tracks, 81 Artists

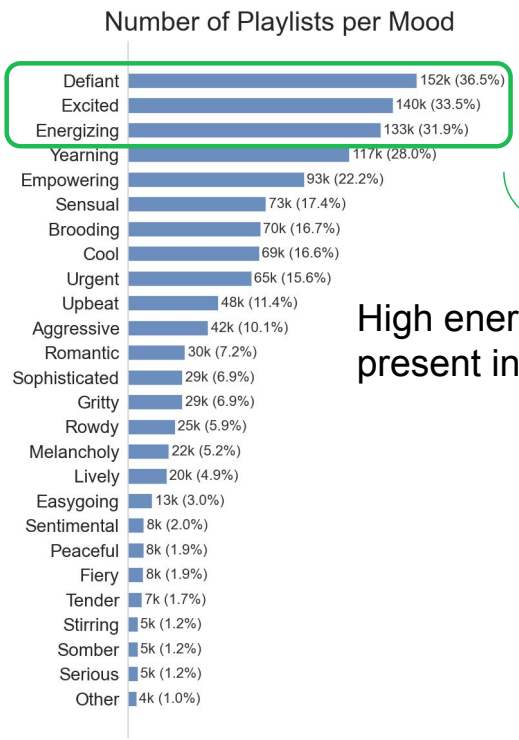
A playlist has 195 tracks and 81 artists on average

Genres & Moods

Certain genres, moods are represented more in the sample.



Pop, dance & house, and indie rock appear in 79% of all playlists

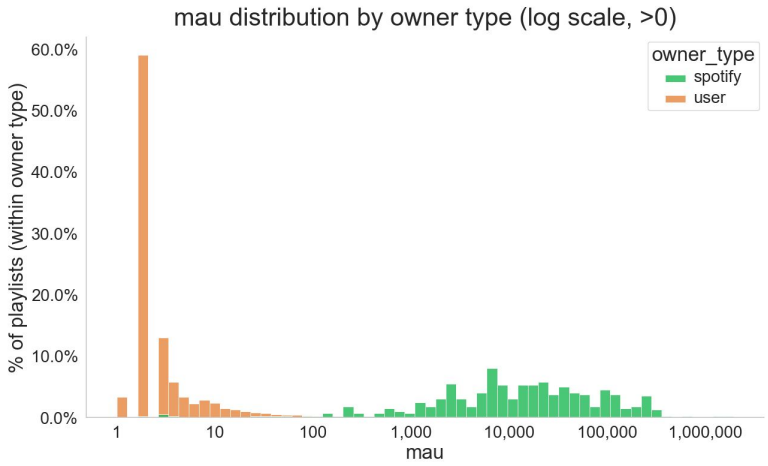


High energy moods are present in 62% of playlists

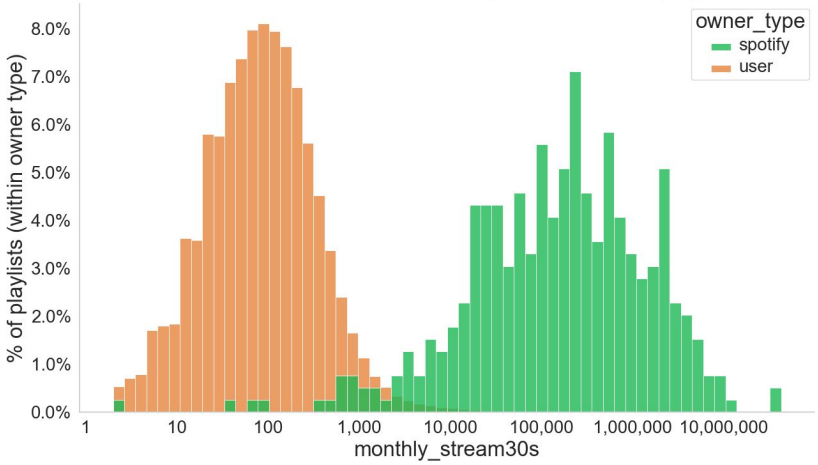
Reach & Activity

Spotify playlists skew towards higher monthly active users and monthly streams

MAU distribution by owner type



Monthly Streams distribution by owner type

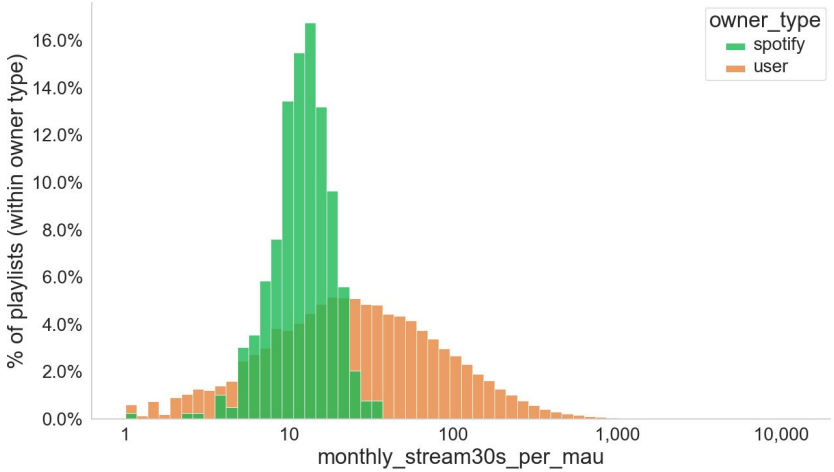


Engagement per user & Retention

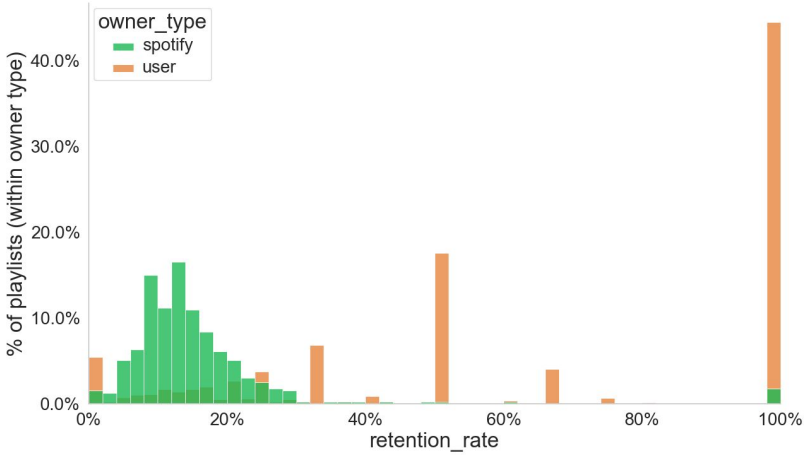
The story changes when we look at engagement per user (Monthly Streams / MAU) and retention (MoM Retention)

User playlists have higher median engagement per user and retention

Engagement distribution by owner type



Retention distribution by owner type



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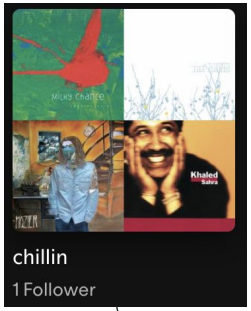
Success Definition

Starting point: Reach and activity metrics (monthly active users, monthly streams) are natural first candidates for defining success. More listeners, more streams, more successful.

Concern: This reduces success to popularity. A Spotify playlist with millions of listeners will always win on reach.



me :)

Success Definition

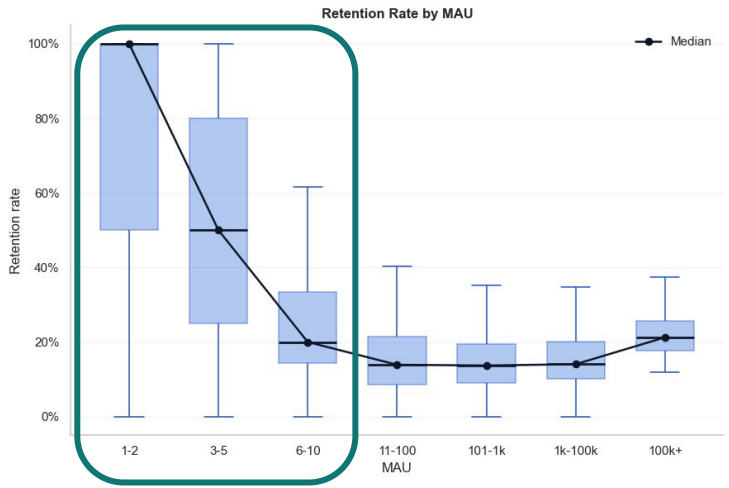
Ultimately a successful playlist should help Spotify's value add to a user; spend more time on the platform and bring them back to Spotify

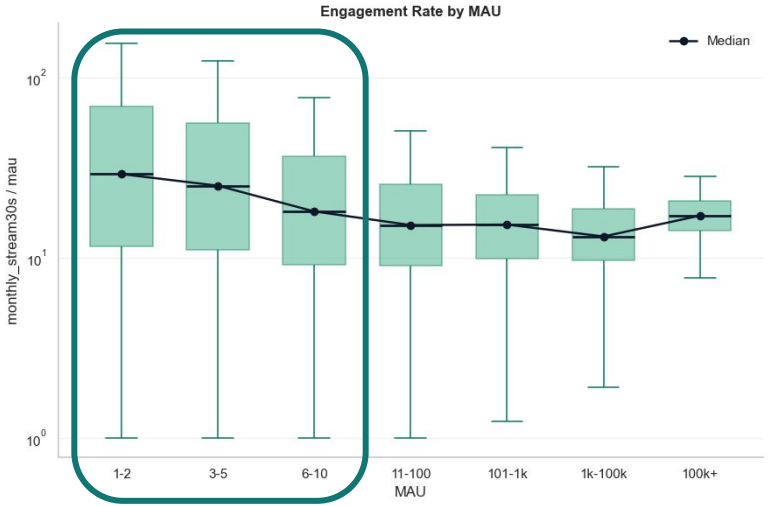
- **Per user engagement**: We measure this as monthly streams per user. This tells us how deeply each user interacts with the playlist
  - (Monthly Streams / MAU)
- **Retention**: This tells us what fraction of last month's users found enough value to return
  - (MoM Retention)

- **Criteria**: having above median per user engagement **AND** retention

Success Definition

Popular playlists behave differently compared to personal playlists in retention and engagement; using the same thresholds would not be fair





Proposed Approach:

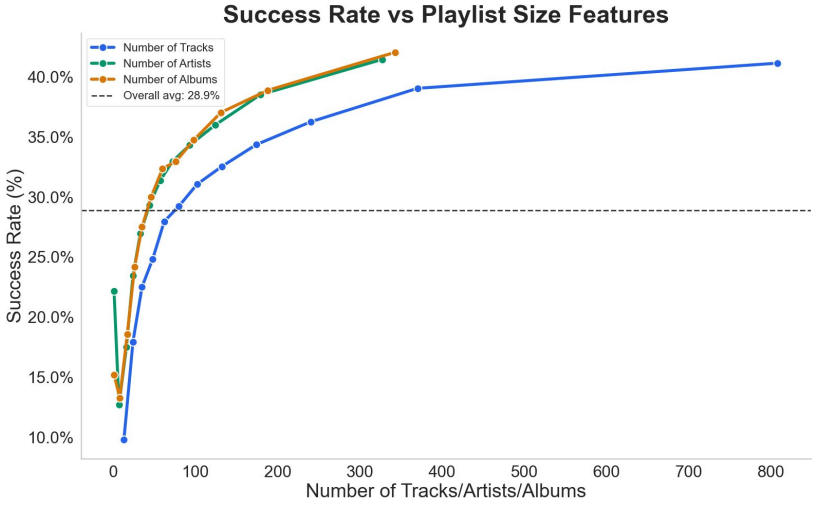
Success is defined as above median per user engagement AND retention; median is calculated separately for low MAU and higher MAU playlists to ensure a fair comparison

	Median Engagement (Streams per MAU)	Median Retention (Monthly)	% Successful
Personal Playlists MAU (1-10)	>27	>75%	**28%**
Popular Playlists MAU (11+)	>15	>14%	**34%**
ALL	-	-	**29%**

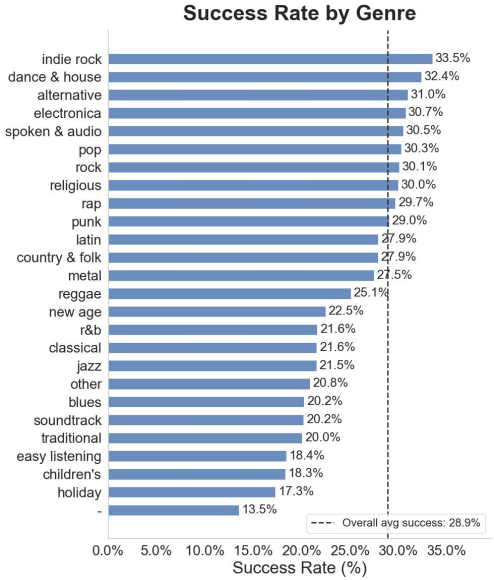
Deep Dive into Success

Playlists that are more diverse, have more tracks are more often successful according to our definition

Success appears to be correlated with playlist diversity and size



Indie Rock, Dance & House, Alternative playlists are more often successful



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Modelling Approach

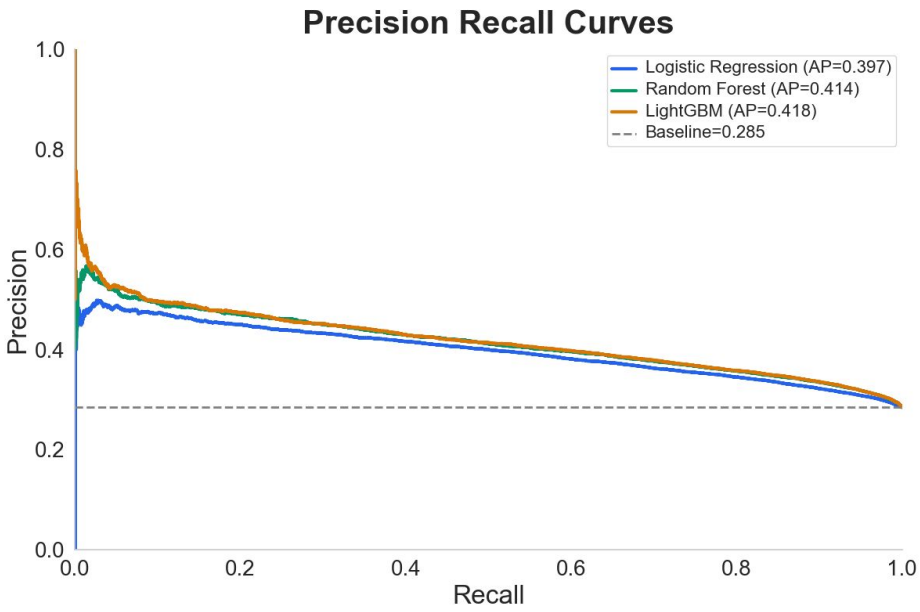
A predictive model is built to understand how much each factor explain success

Features Used:
Playlist Size and Diversity (number of tracks, number of artists, number of albums)
Track Freshness (number of local tracks, % of local tracks, is positive local tracks)
Concentration Ratios (track per album, track per artist, album per artist)
Title Token Count
Title Keyword Flags (new, chill, workout, love, etc.)
Playlist Purpose Inferred from Title (workout, worship, party, study_focus, sleep_relax, discovery_fresh)
Genre Tags (is pop, rap, …)
Mood Tags (is excited, is peaceful, …)
Owner Type

Additional features are created to be used in the model

Modelling Approach

A predictive model is built to understand how much each factor explain success



LightGBM leads with highest PR-AUC Score

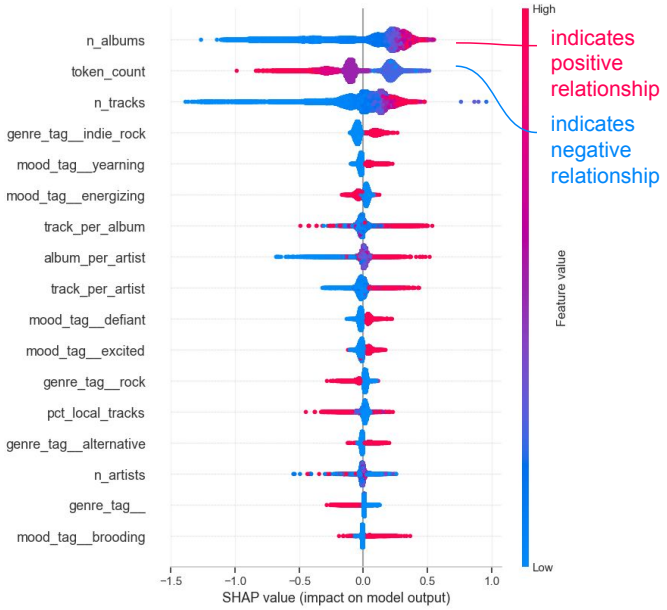
What this means?

A model with no real knowledge of what makes a playlist successful would score 0.29. Our best model scores 0.42 meaning **it's nearly 50% better at identifying successful playlists than random chance**.

Modelling Approach

A predictive model is built to understand how much each factor explain success

Feature Importance



Takeaways

Catalog Diversity: The strongest predictor of playlist success is catalog breadth. Playlists that include a wider range of unique albums and tracks are more likely to be classified as successful

Title Length: Token count is the second most important feature, and it works inversely: playlists with shorter, more focused titles are pushed toward success

Concentration Ratios: Track per artist, track per album, and album per artist all rank in the top 10 importance. Having more tracks and albums from an artist is associated with success

Genre and Mood signals: Among genres, indie rock pushes predictions toward success when present. On the mood side, yearning, defiant, and excited show similar positive effects. Playlists with no identifiable genre are penalized

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Summary

Spotify vs User Playlists

Spotify playlists tend to outperform user playlists in terms of active users, while, on average user playlists win in per-user engagement and retention rate, indicating their ability for building a deeper connection with listeners, a necessary part of Spotify's value.

Success Criteria

Playlists with low vs high MAU show different engagement and retention patterns. Applying different success thresholds for each group ensures a fair comparison.

Success Metrics

MAU and number of streams are important metrics, but they are highly dependent on distribution. A successful playlist should make users listen more and come back more often, so per-user engagement and retention are chosen for defining success.

Success Predictors

Catalog diversity and title conciseness emerge as the strongest predictors of success; however, this does not indicate causality.

Next Steps

Longer timeframe data

Long term success of the playlist can be determined with longer timeframe (e.g. YoY) retention & growth metrics

Further playlist categorization

Playlists with different purposes can be benchmarked within its own category for even a better success definition

More metrics

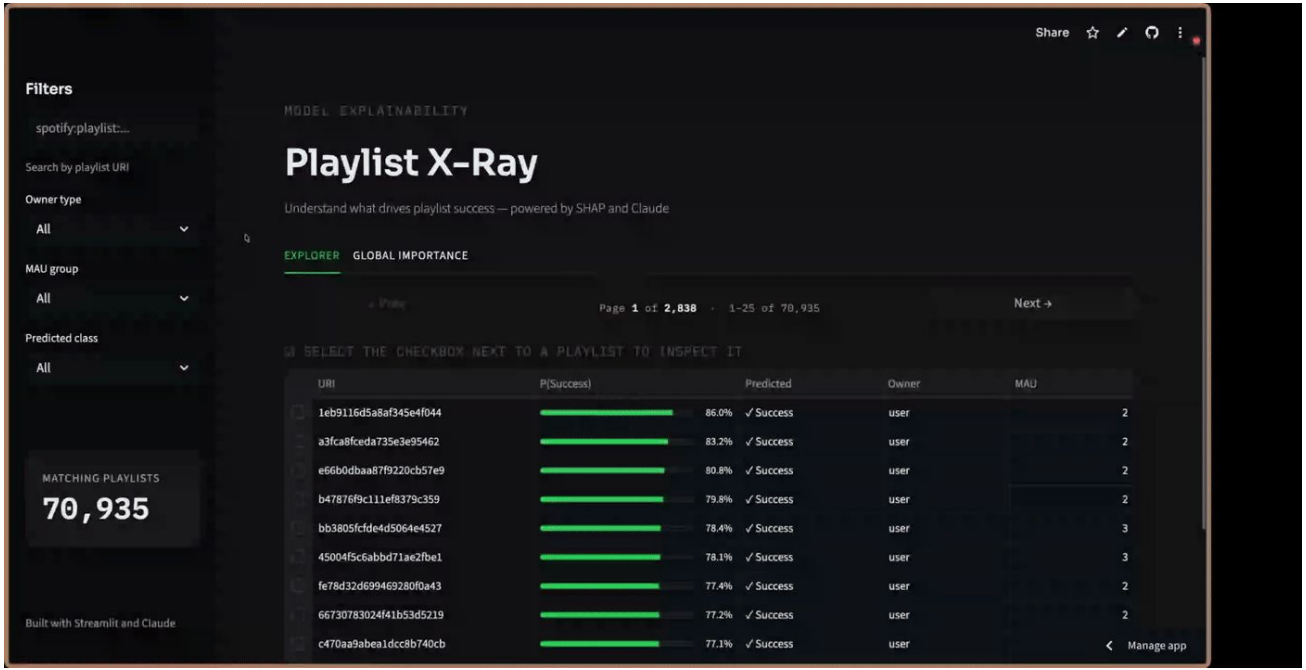
Other potential success metrics and potential predictive features are

- Discovery rate
- Share of user streams coming from a playlist
- User metadata
- Playlist history
- Track level data

Experimentation

We can design experiments by changing one thing at a time in the playlist to understand the true causality

Quick DEMO: Playlist X-ray Dashboard



Link: Streamlit

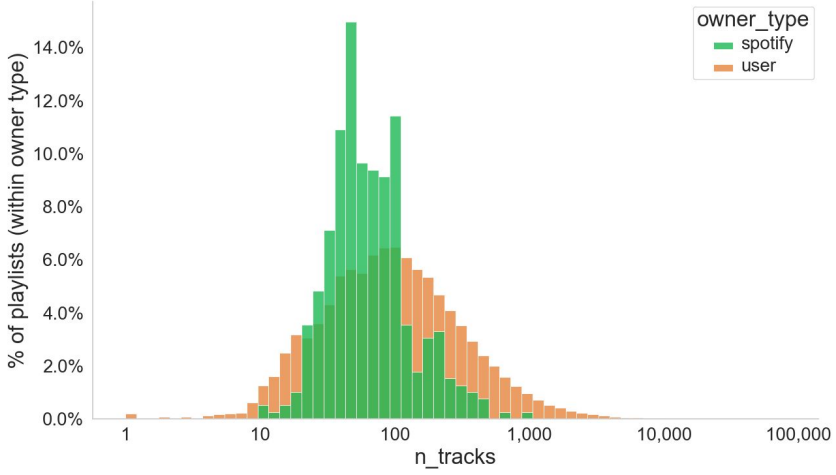
Q & A

Appendix

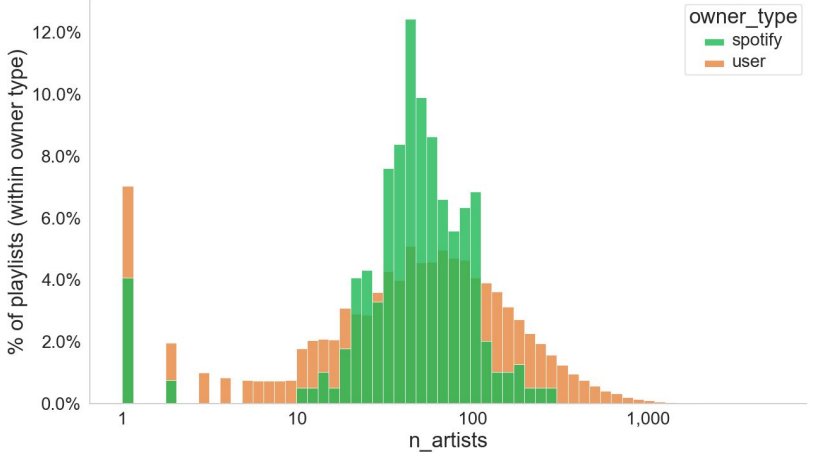
Track and Artist Variance

User playlists have wider range of number of tracks and artists compared to Spotify playlists

Number of tracks distribution by owner type

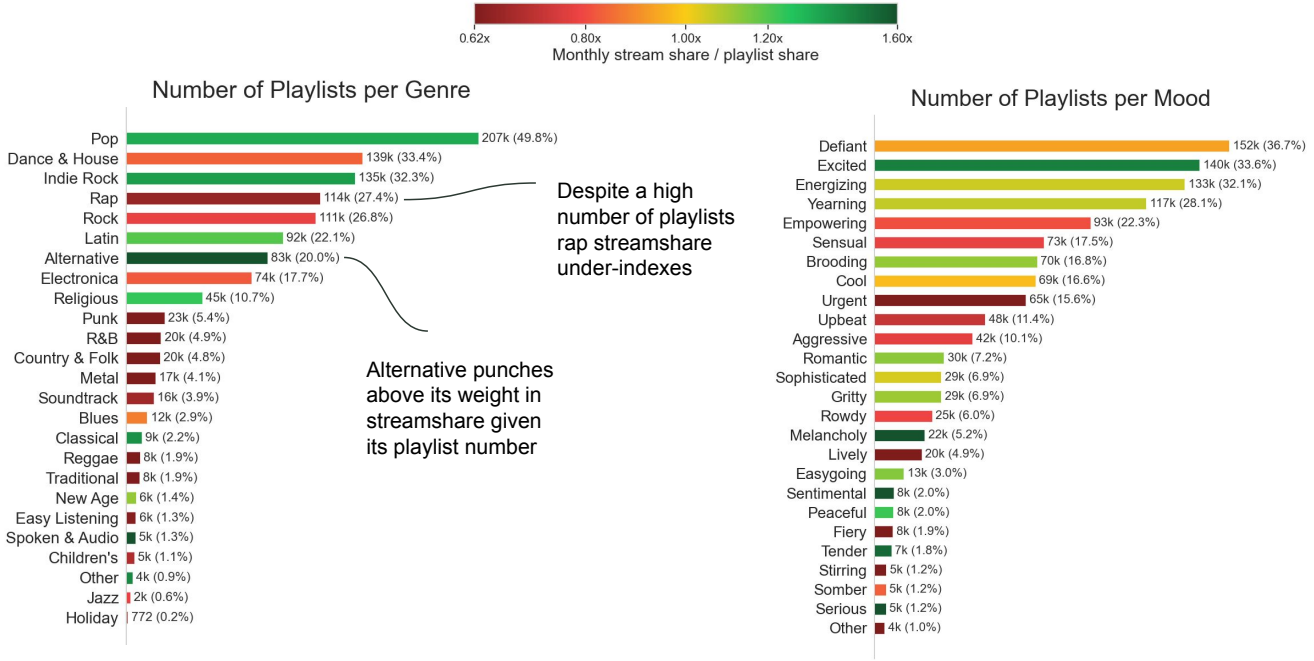


Number of artist distribution by owner type



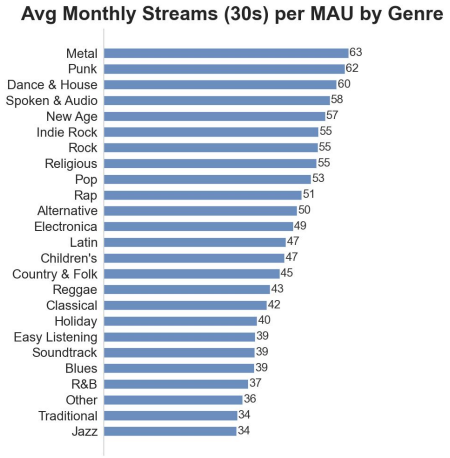
Genre & Mood Playlist and Stream Shares

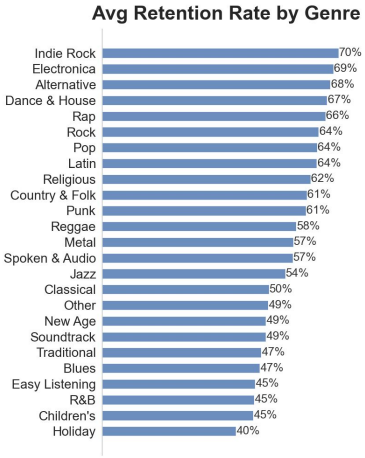
Pop, Latin and Alternative punches above their weight. They have higher stream share compared to their playlist share. Moods have a more balanced playlist and stream shares compared to Genres.

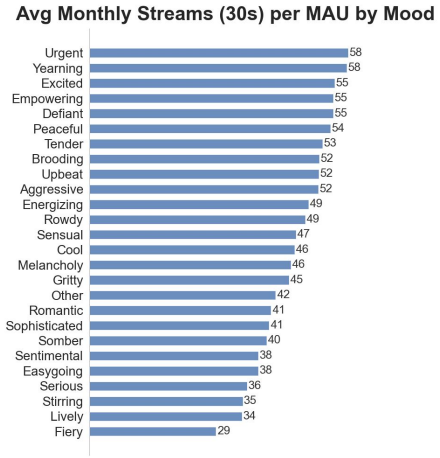


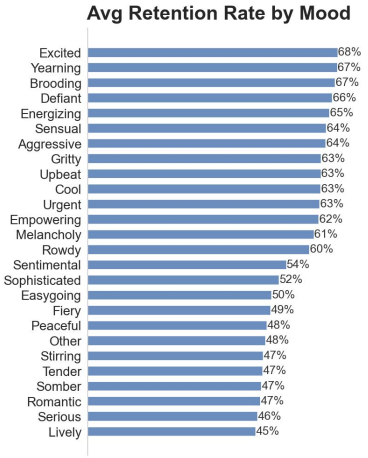
Genre & Mood Success Metrics

Per user engagement and retention rates for each genre and mood.



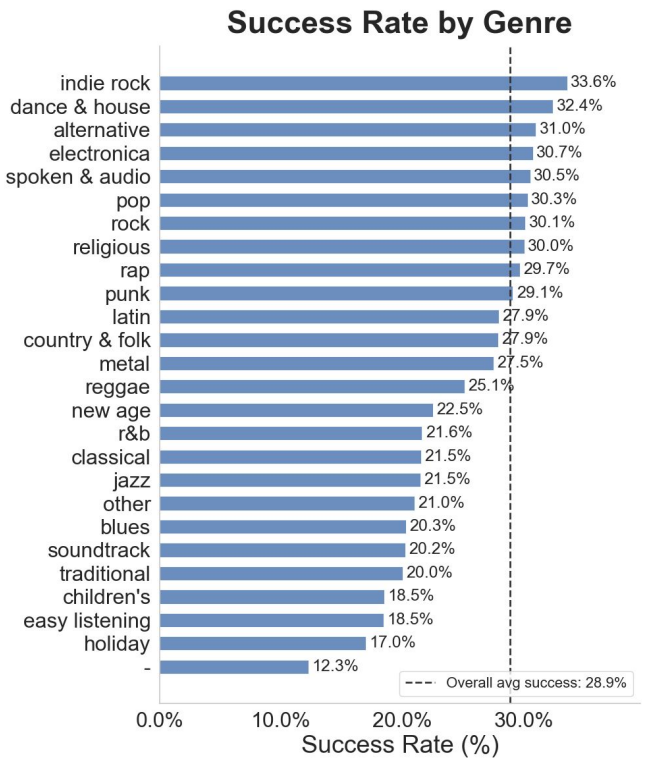


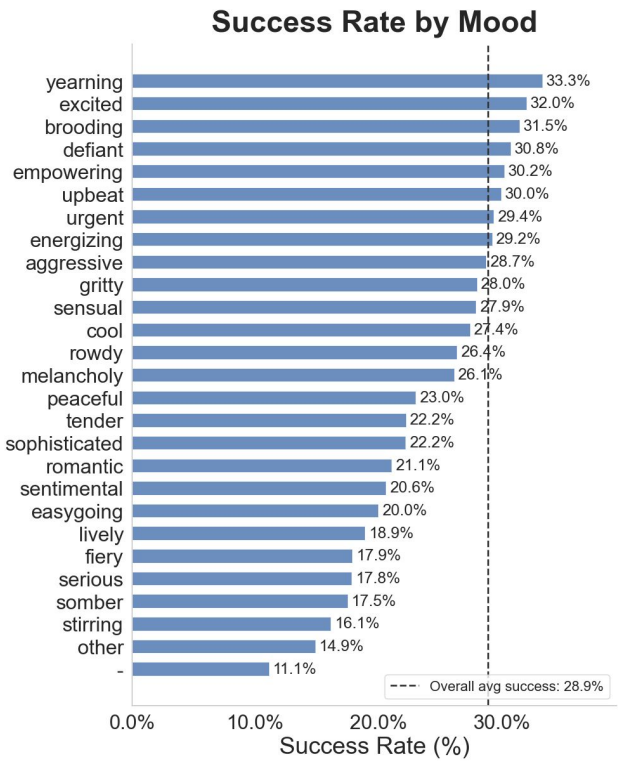


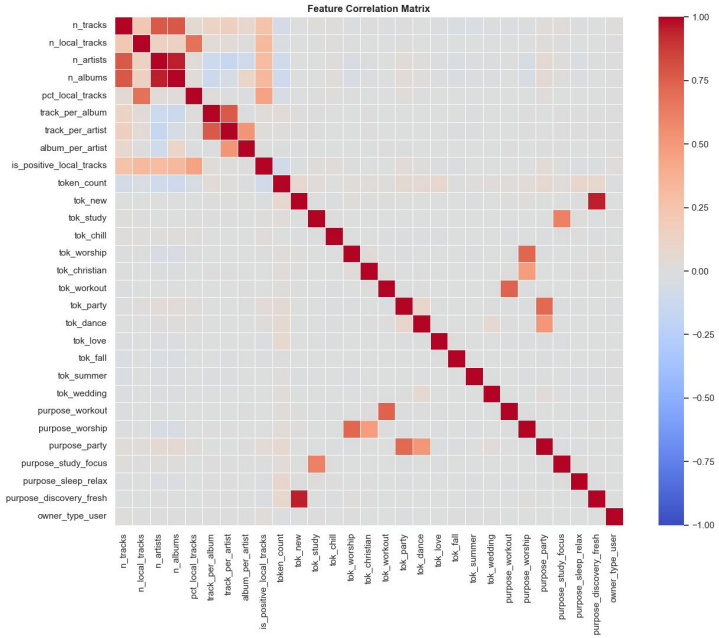


Deep Dive into Success

Success % varies across Genres and Moods





Feature Correlations

Certain features are highly correlated with each other: Number of tracks, albums and artists show strong positive correlation